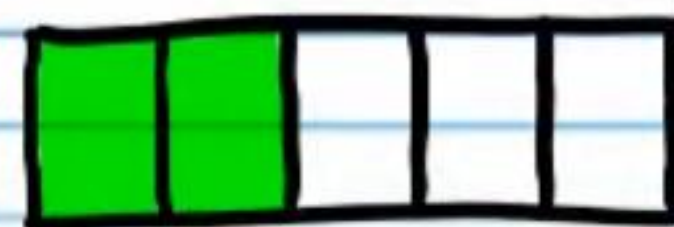
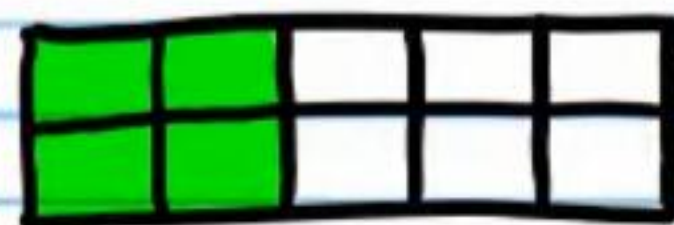


Class 11: Fractions and Fraction Equivalency

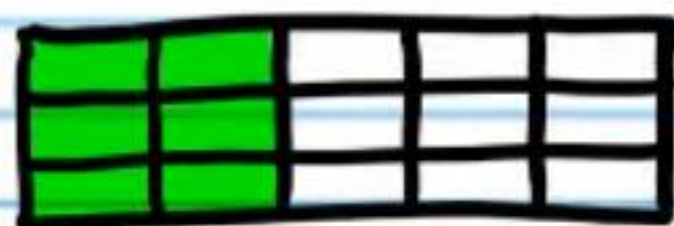
$\frac{2}{5}$ ← numerator
← Denominator



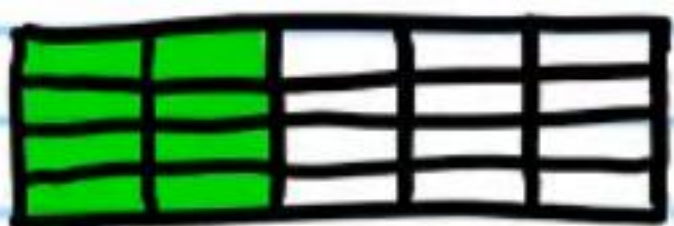
$$\frac{2}{5} = \frac{2 \times 2}{5 \times 2} = \frac{4}{10}$$



$$\frac{2}{5} = \frac{2 \times 3}{5 \times 3} = \frac{6}{15}$$



$$\frac{2}{5} = \frac{2 \times 4}{5 \times 4} = \frac{8}{20}$$



$$\frac{1}{2} = \frac{1 \times 2}{2 \times 2} = \frac{2}{4}$$

$$\frac{1}{2} = \frac{1 \times 3}{2 \times 3} = \frac{3}{6}$$

$$\frac{1}{2} = \frac{1 \times 4}{2 \times 4} = \frac{4}{8}$$

$$\frac{40}{60} = \frac{40 \div 2}{60 \div 2} = \frac{20}{30}$$

$$\frac{40}{60} = \frac{40 \div 10}{60 \div 10} = \frac{4}{6}$$

$$\frac{40}{60} = \frac{40 \div 20}{60 \div 20} = \frac{2}{3}$$

① Find three fractions that are equivalent to $\frac{3}{4}$ by using multiplication.

② Find three fractions that are equivalent to $\frac{8}{9}$ by using multiplication.

③ Find three fractions that are equivalent to $\frac{4}{11}$ by using multiplication.

④ Find three fractions that are equivalent to $\frac{100}{300}$ by using division.

⑤ Find three fractions that are equivalent to $\frac{200}{700}$ by using division.

⑥ Find three fractions that are equivalent to $\frac{16}{24}$ by using division.

⑦ Find three fractions that are equivalent to $1\frac{3}{48}$ by using division.

Reduce the following fractions to lowest terms.

⑧ $\frac{32}{40}$

⑮ $\frac{48}{72}$

⑨ $\frac{72}{81}$

⑯ $\frac{50}{60}$

⑩ $\frac{8}{10}$

⑰ $\frac{9}{30}$

⑪ $\frac{56}{64}$

⑱ $\frac{12}{20}$

⑫ $\frac{35}{49}$

⑲ $\frac{5}{45}$

⑬ $\frac{7}{18}$

⑳ $\frac{100}{500}$

⑭ $\frac{15}{25}$

㉑ $\frac{44}{77}$

㉒ $\frac{18}{27}$

㉓ $\frac{24}{54}$

㉔ $\frac{10}{19}$

㉕ $\frac{10}{24}$

㉖ $\frac{28}{42}$

㉗ $\frac{20}{30}$

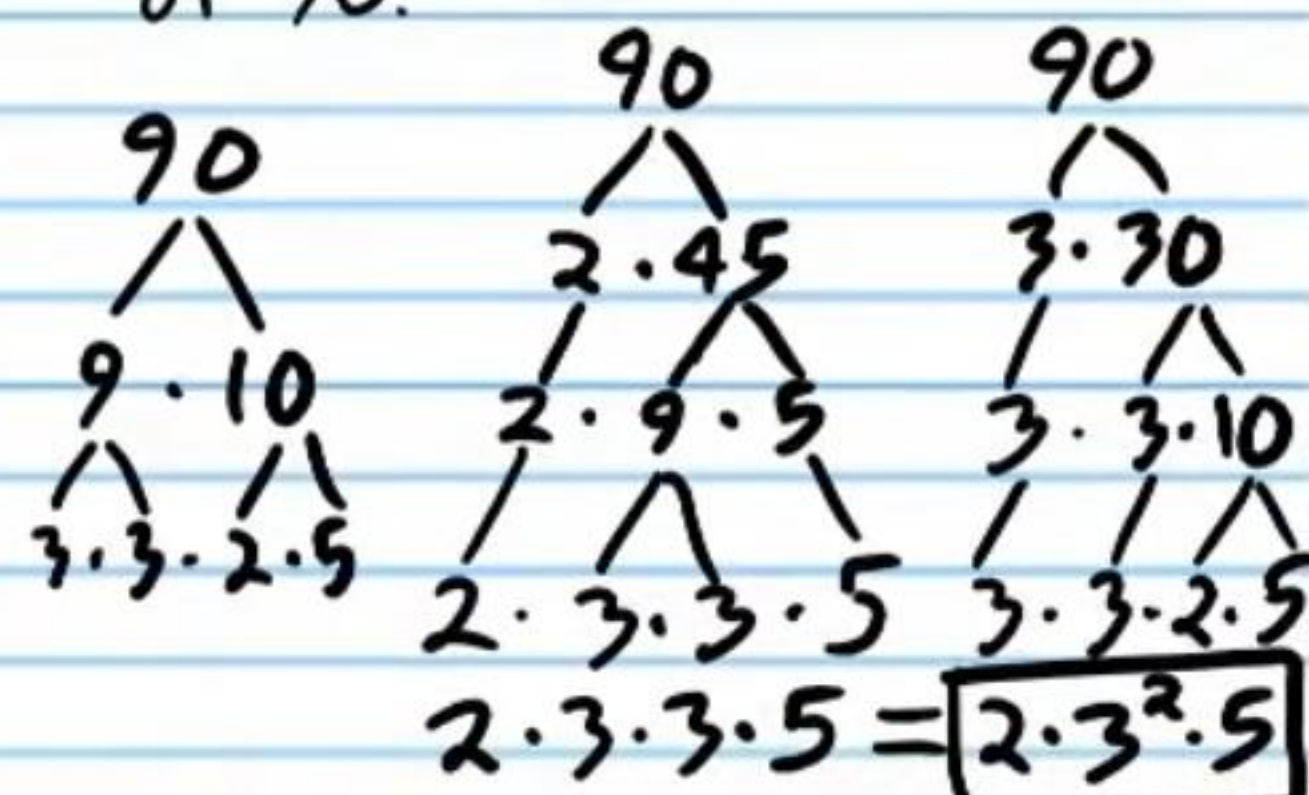
㉘ $\frac{81}{90}$

㉙ $\frac{28}{64}$

㉚ $\frac{4}{48}$

㉛ $\frac{50}{75}$

㉜ Find the prime factorization of 90.



㉝ Find the prime factorization of 48.

㉞ Find the prime factorization of 50.

㉟ Is 8 composite or prime?

(36) IS 7 composite or prime?

(37) IS 15 composite or prime?

(38) IS 29 composite or prime?

(39) $\frac{75}{120}$

(40) $\frac{4}{80}$

(41) $\frac{6}{72}$

(42) $\frac{6}{81}$

(43) $\frac{45}{100}$

Find the greatest common factor (GCF) of the following pairs of numbers and use it to reduce the corresponding fraction

(44) 40 and 32

④⑤ 90 and 110.

④⑧ 21 karat gold is $\frac{21}{24}$ gold. What fraction of gold is 21 karat gold in lowest terms?

④⑥ 124 and 100.

④⑨ 16 out of 18 people at the football stadium bought hot dogs. What fraction of people bought hot dogs in lowest terms?

④⑦ 66 and 88.